FreshRoute - A Smart Fruit Supply Chain System to Reduce Waste and Eliminate Middlemen in Sri Lanka

25-26J-444

Project Proposal Report

Dias N. A.

B.Sc. (Hons) Degree in Information Technology Specializing in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

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DECLARATION

I declare that this is my own work and does not incorporate without acknowledgment any material previously submitted for a degree or diploma at any other University or Institute of Higher Learning. To the best of my knowledge, it does not contain any material previously published or written by another person except where due reference is made.

Name	Student ID	Signature	Date
Dias N.A.	IT22213730		28-08-2025

As the supervisor/s of the above-mentioned candidates, I hereby certify that they are conducting research for their undergraduate dissertation under my guidance and direction.

Signature of the Supervisor

28-08-2025

Date

Signature of the Co-Supervisor

28-08-2025

Date

ABSTRACT

Sri Lanka's fruit supply chain is characterized by systemic inefficiencies, including high post-harvest losses, opaque market structures, and the dominance of intermediaries who suppress farmer income and inflate consumer prices. This research proposes a foundational solution to disintermediate this inefficient system by leveraging blockchain technology to create a transparent and trusted marketplace. The study will design, develop, and evaluate a decentralized platform that facilitates direct trade between farmers and buyers. The core novelty lies in the creation of a verifiable, on-chain reputation system for farmers, built upon Decentralized Identifiers (DIDs). This "digital social capital" will serve as the basis for an intelligent matching system, connecting buyers with reliable, high-quality producers. The anticipated outcome is a significant reduction in the influence of intermediaries, leading to fairer prices, timely payments for farmers, and enhanced traceability for buyers. The methodology will employ a mixed-methods approach, combining qualitative stakeholder analysis with the development of a Hyperledger Fabric-based prototype, which will be validated through a targeted pilot study. This research aims to demonstrate the viability of a blockchain-powered marketplace as a practical and transformative tool for building a more resilient, equitable, and efficient agricultural ecosystem in Sri Lanka.

Keywords: Blockchain, Supply Chain, Disintermediation, Smart Contracts, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
B2B	Business-to-Business
CA	Certificate Authority
DEC	Dedicated Economic Center
DID	Decentralized Identifier
DSR	Design Science Research
IoT	Internet of Things
IPFS	Inter Planetary File System
KPI	Key Performance Indicator
PHL	Post-Harvest Loss
VC	Verifiable Credential
WBS	Work Breakdown Structure

1 INTRODUCTION

1.1 Background Literature

The agricultural sector remains a cornerstone of the Sri Lankan economy, providing livelihoods for a significant portion of the rural population.[1] The fruit sub-sector, in particular, holds immense potential for economic growth, food security, and export revenue. However, this potential is severely undermined by deeply entrenched systemic inefficiencies that have persisted for decades.[1] The traditional fruit supply chain is characterized by high post-harvest losses (PHL), opaque market structures, and the dominance of intermediaries who suppress farm-gate prices while inflating consumer costs.[2]

This long-standing issue has been critically exacerbated by the recent national economic crisis, which has driven up input costs, disrupted logistics, and increased the financial vulnerability of smallholder farmers, making systemic reform an urgent national priority.[3] These challenges highlight the need for a foundational shift in how the supply chain operates, moving away from a system reliant on informal networks toward one that is transparent, efficient, and equitable.

This research proposes to address these challenges by designing and evaluating a blockchain-powered direct marketplace. The core of this solution is the creation of a trusted ecosystem where farmers and buyers can transact directly, guided by a system of verifiable, on-chain reputation. By replacing the informal, often exploitative, trust networks of intermediaries with a transparent and immutable digital ledger, this project aims to fundamentally re-engineer the market dynamics to favor efficiency, quality, and fair value for all participants.

1.2 Literature survey

1.2.1 The State of Sri Lanka’s Fruit Supply Chain

The most glaring symptom of the supply chain's dysfunction is the staggering level of post-harvest losses (PHL). A 2018 report from the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI)[4] found that post-harvest losses for fruits varied between 20-40%, with highly perishable produce like papaya experiencing losses as high as 46%. For vegetables, the losses ranged from 20-46%, with okra being the most affected. These figures are corroborated by a report from the Auditor General's Department [5], which points to systemic failures in post-harvest management.

Table 1: Post-Harvest Losses in Sri Lankan Fruit and Vegetable Sector

Category	Post-Harvest Loss (%)	Key Affected Produce	Primary Causes	Source
Fruits	20 - 40%	Papaya, Mango, Banana, Pineapple	Mechanical damage, rapid ripening, diseases, poor handling	[4]
Vegetables	20 - 46%	Okra, Tomato, Carrot	Unsatisfactory packaging, lack of ventilation, poor handling	[4]
Transportation	19 - 21%	All Fruits & Vegetables	Improper transport methods, lack of cold chain	[6]

These losses are driven by a combination of factors. Gunarathna and Bandara [2] identify several key logistical bottlenecks. Their research highlights that the supply chain is overwhelmingly dependent on road transport, often in non-refrigerated, open-sided lorries where overloading and poor road conditions lead to significant compression damage and bruising. The World Bank [7] further notes that around 97% of domestic freight is transported by road, a situation that creates congestion and increases costs. The recent economic crisis has severely compounded these issues, with a HARTI study [3] revealing that lorry fees increased fivefold, drastically raising the cost of transport.

Inadequate packaging and storage are also major contributors. Gunarathna and Bandara [2] point to the prevalent use of cheap polysacks, which offer no protection and accelerate spoilage. While modern supermarket chains have successfully reduced their PHL to as low as 2-6% by using plastic crates and refrigerated trucks, this infrastructure is over 50% more expensive and remains inaccessible to most independent farmers. This is compounded by a severe lack of cold storage facilities near production zones, a weakness identified by the Food and Agriculture

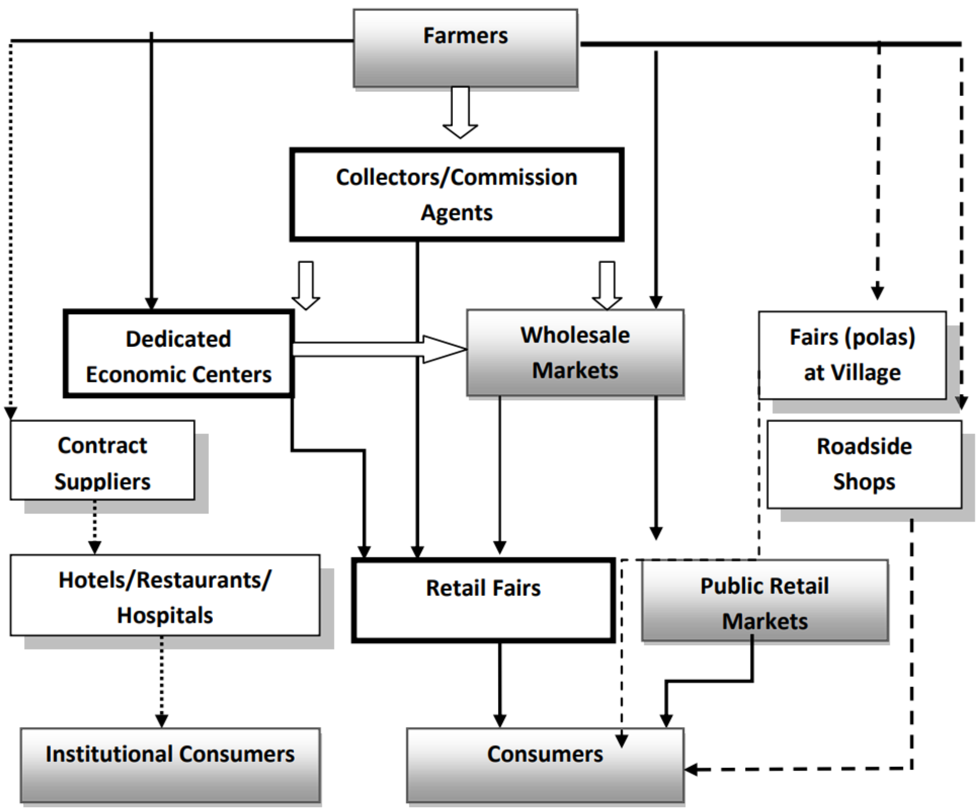
Organization (FAO) [8], which forces farmers into immediate post-harvest sales, leading to market gluts and price crashes.

A fundamental flaw in the traditional system, as noted by Gunarathna and Bandara [2], is that farmers are paid by weight, not by quality grade. This creates a perverse incentive structure where there is no financial reward for careful handling or investing in better practices, and it can even encourage malpractices like concealing low-quality produce to inflate weight.

1.2.2 The Intermediary Challenge and the Failure of Economic Centers

The traditional fruit and vegetable supply chain, which handles over 90% of produce, is controlled by a complex network of intermediaries.[2] While these actors provide necessary functions like aggregation and distribution, their unchecked dominance is a primary driver of economic inefficiency and inequity. The power of these intermediaries stems from several sources. Gunarathna and Bandara [2] highlight the critical issue of information asymmetry, where farmers in rural areas lack access to real-time market prices, making them dependent on local collectors. Their study shows that the cumulative margin for intermediaries can be as high as 41-49% of the final retail price. For example, carrots bought from farmers at Rs. 120 can sell for as much as Rs. 400 in urban markets, with the difference absorbed by middlemen.

A study on conflicts in the agriculture sector by Herath and Siriwardhane [9] further explains that a critical and often overlooked function of intermediaries is that of informal creditors. By providing pre-season advances for inputs, they create debt obligations that "lock-in" farmers, forcing them to sell their entire harvest exclusively to that intermediary at a disadvantageous price. Finally, as individual smallholders lack the capacity to transport small harvests to distant markets, they rely on intermediaries for logistical control to achieve the necessary economies of scale.[8]



Source: Vidanapathirana, Priyadarshana and Rambukwella, 2011

Figure 1:The Traditional Sri Lankan Fruit and Vegetable Supply Chain

Established in 1999, the 19 Dedicated Economic Centers (DECs) were intended to solve this problem by creating open wholesale marketplaces. However, this initiative has largely failed. Instead of empowering farmers, DECs have been captured by powerful traders and commission agents who operate as a monopoly, manipulate prices, and contribute to immense wastage within the centers themselves.

Table 2: Cost Factor Contribution to Final Vegetable Price in Sri Lanka

Cost Factor	Contribution to Final Price (%)	Implication	Source
Profit (All actors)	33 - 44%	A significant portion of the final price is profit, much of which is captured by intermediaries.	[2]
Post-Harvest Loss	9 - 33%	The cost of wasted produce is passed on to the consumer, inflating the price of the remaining goods.	[6]
Commission	5 - 10%	Direct fees charged by commission agents at wholesale markets.	[2]
Transport	1 - 3%	While a smaller percentage, this cost has surged post-crisis, impacting both farmer and consumer prices.	[3]

1.2.3 Existing Digital Solutions and Government Initiatives

The Sri Lankan government, often in partnership with international bodies, has launched several key initiatives aimed at modernizing the agricultural sector. While these efforts have made progress, they also highlight the remaining gaps that new technological solutions aim to fill.

Government-Led Modernization Efforts

Two of the most significant government-led programs are the Agriculture Sector Modernization Project (ASMP) and the ongoing work of the Ministry of Agriculture's Technology Division.

Agriculture Sector Modernization Project (ASMP) is a major project designed to shift the agricultural sector towards a more commercial and export-oriented model. Its primary goals are to improve market linkages, encourage agricultural diversification, and promote the adoption of new technologies. The ASMP provides financial incentives, including bank guarantees for loans, to entrepreneurs and farmer organizations willing to invest in high-value production and value-addition. The project covers a wide range of fields, from fruits, vegetables, and spices to the development of new technological equipment and IT software systems for the agricultural sector. The core concept is to build partnerships between public and private entities to solve bottlenecks along the entire value chain, from the farmer to the consumer.[5]

Ministry of Agriculture's Technology Division serves as the operational arm for implementing technology-focused policies. Its responsibilities are broad and include sponsoring research projects, regulating the import and export of food crops and agro-inputs, developing projects with the private sector, and introducing appropriate agricultural machinery. A key focus is on promoting eco-friendly "Green Agriculture" systems by improving crop varieties and managing inputs like nutrients and plant protectors. This division essentially provides the legal and regulatory framework needed to support the modernization goals of projects like the ASMP.[10]

Collaboration with International Partners

The Food and Agriculture Organization of the United Nations (FAO) has been a long-standing partner in Sri Lanka's agricultural development, with its role becoming particularly crucial during and after the recent economic crisis. Beyond emergency aid, the FAO is involved in identifying core weaknesses in the supply chain. A major food flow mapping exercise conducted in 2024 highlighted critical issues in transportation, storage, and rural-urban connectivity that contribute to food losses and system fragility. This work provides an empirical basis for targeted interventions aimed at building a more resilient food system.[13]

Emerging Digital Platforms and Localized Markets

Alongside government efforts, a nascent agri-tech ecosystem has begun to emerge in Sri Lanka, producing several innovative but often localized solutions. While direct farmer-to-consumer markets like "Good Market" have shown success in niche urban areas, they have not been able to scale sufficiently to disrupt the mainstream, intermediary-dominated system.

More recently, several tech startups have started to tackle different parts of the supply chain:

- **GoviLab and Sarvodaya's Fusion ICT4D:** These initiatives have introduced mobile apps and market information platforms to improve farmers' access to critical data.
- **SenzAgro:** This company provides a "smart farming" platform that uses real-time data to help smallholder farmers enhance crop productivity and optimize the use of water and fertilizer.

- **Agrithmics (now Cultive8):** This startup developed an ERP solution focused on agriculture, successfully helping plantation companies and tea factories digitize their procurement and payment processes.
- **Agroworld Web:** This platform aims to be an "investment-to-shelf" digital solution, connecting farmers and investors to optimize the agricultural value chain.

While these platforms demonstrate the potential of technology, they tend to be specialized in specific sectors (like tea) or focused on particular aspects of farming (like crop intelligence). A comprehensive, nationwide digital marketplace that addresses the fundamental issue of trust and transparency in B2B transactions between the vast network of smallholder farmers and large-scale buyers remains a critical missing piece.

1.2.4 International Precedents and Direct Sourcing Models

Globally, the shift towards direct sourcing models represents a significant trend in modern supply chain management. This move is driven by large retailers and buyers who are seeking to enhance efficiency, improve traceability, and achieve greater coordination within their supply chains. A systematic review by Chen, Liu, and Wang [14] confirms the extensive body of research exploring how blockchain technology can facilitate this shift by creating more transparent and resilient food supply chains.

The application of blockchain to achieve these goals has been successfully demonstrated in several high-profile international case studies. The most notable of these is IBM Food Trust, a platform built on the Hyperledger Fabric framework, which is now used by global giants like Walmart,[15] Carrefour, and Nestlé. In a landmark pilot study, Walmart leveraged the platform to trace the origin of mangoes sold in its US stores. The time required to perform this task was dramatically reduced from a full seven days to a mere 2.2 seconds. This case powerfully demonstrates the technology's capacity to provide rapid, reliable traceability, a function that is critical for managing food safety, executing swift product recalls, and verifying the authenticity of products. Similarly, other platforms like Australia's AgriDigital have successfully applied blockchain to bring new levels of efficiency and trust to the grain supply chain,[16] showcasing the technology's versatility across different agricultural commodities.

These international examples provide compelling proof that blockchain is a viable and transformative technology for agri-food supply chains. However, it is crucial to recognize that these are often "top-down" initiatives, driven and funded by large, multinational corporations with immense resources. Their success is predicated on the ability to mandate participation from their suppliers. The challenge, and the central focus of this research, is to design a system that is accessible, affordable, and empowering for the fragmented base of smallholder farmers in a developing country context like Sri Lanka, where a top-down corporate mandate is not feasible.

1.2.5 Blockchain as a Foundational Technology for Trust

The core problem this research addresses is the profound trust deficit that pervades the Sri Lankan fruit supply chain. Farmers do not trust that they will receive fair prices or timely payments from unknown buyers, while buyers do not trust the quality, origin, or reliability of produce sourced from a fragmented network of smallholders. Blockchain technology is uniquely suited to solving this problem by creating a shared, immutable, and transparent "single source of truth" that is not controlled by any single entity.

Key Technological Components:

- **Permissioned Blockchain (Hyperledger Fabric):** For an enterprise-grade, business-to-business (B2B) network, a permissioned blockchain is essential. Hyperledger Fabric, an open-source platform hosted by the Linux Foundation, is the industry standard for such applications.[17] Unlike public blockchains where anyone can participate, a permissioned network ensures that only verified and authorized participants (e.g., vetted farmers, registered buyers, delivery driver) can join. This is critical for creating a secure business environment. Furthermore, Fabric's "channel" architecture allows for the creation of private sub-networks where confidential transactions can occur.

- **Smart Contracts:** These are self-executing digital agreements where the terms of the trade are written directly into code on the blockchain. In the proposed system, a "Smart Sales Agreement" can automate the entire transaction lifecycle. For instance, the smart contract can be programmed to lock a buyer's payment in a secure escrow account at the time of the agreement. It will then automatically release the funds to the farmer's digital wallet the moment a verifiable event occurs, such as a trusted quality assessor confirming the successful delivery of the produce. This mechanism effectively eliminates payment delays and counterparty risk, two of the most significant pain points for farmers.

- **Decentralized Identity (DID) and Verifiable Reputation:** The core novelty of the proposed solution is the integration of Decentralized Identity (DID) and Verifiable Credentials (VCs). Müller and Schmidt [18] explore the use of DIDs for agri-food supply chains, arguing they can create a form of "digital social capital." Similarly, Perera and Silva [19] discuss leveraging DIDs and IoT for transparent farm-to-fork supply chains. Each farmer can be issued a DID, a self-sovereign digital identity they own and control. Every successful transaction is recorded as a Verifiable Credential (VC) linked to their DID, creating an immutable track record of their performance. This verifiable reputation becomes the new basis for trust, allowing buyers to confidently engage with farmers they have never met, thereby breaking the information monopoly of traditional intermediaries.

Table 3: Comparison of Blockchain Platforms for Enterprise Supply Chains

Feature	Hyperledger Fabric	Solana	Polygon
Ledger Type	Permissioned	Permissionless	Permissionless (PoS)
Data Privacy	High (Channels, Private Data Collections)	Low (Public Ledger)	Low (Public Ledger)
Governance	Consortium-based	Foundation-based	Foundation-based
Suitability for B2B	**Excellent.** Designed for enterprise use with a focus on privacy, confidentiality, and accountability between known participants.	**Poor.** Public nature and lack of granular privacy controls are unsuitable for sensitive commercial data like pricing.	**Poor.** Primarily designed for public dApps; lacks the inherent privacy and governance structures needed for B2B supply chains.
Source	[20]	[21]	[22]

1.3 Research Gap

The existing body of literature extensively covers the inefficiencies within Sri Lanka's agri-food supply chain, including post-harvest losses [2] and the dominant role of intermediaries.[9] Separately, a growing number of studies explore the application of blockchain technology for enhancing traceability and transparency in global supply chains.[14]

However, a significant research gap exists at the intersection of these fields, particularly within the unique socio-economic context of Sri Lanka. While the benefits of disintermediation are widely discussed, there is a lack of research on a practical, technology-driven framework to achieve it.

Specifically, there is a lack of research that:

1. Designs and evaluates a blockchain-based direct marketplace tailored to the specific challenges of Sri Lankan farmers, focusing on usability and adoption in a low-digital-literacy environment.[9]
2. Integrates a verifiable, on-chain reputation system using DIDs and VCs as the core mechanism for building trust, a concept explored theoretically by researchers like Müller and Schmidt [18] and Perera and Silva [19], to replace the informal trust networks managed by intermediaries.
3. Proposes a holistic system that combines traceability, reputation, and automated smart contracts to create a complete, end-to-end alternative to the traditional intermediary-led model.

This research aims to fill this gap by focusing on the foundational element of trust. It will demonstrate how a verifiable digital reputation, built on an immutable blockchain ledger, can empower farmers and create the trusted environment necessary for a direct, efficient, and equitable marketplace to thrive.

1.3.1 Research Questions

To address the identified research gap, this study will be guided by the following primary research questions:

RQ1: How can a blockchain-based direct marketplace be designed to be usable and adoptable by smallholder farmers in Sri Lanka, considering the prevalent challenges of low digital literacy and limited access to technology?

RQ2: To what extent can a verifiable on-chain reputation system, built using Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs), effectively replace the socio-economic trust functions of traditional intermediaries in the fruit supply chain?

RQ3: What is the optimal system architecture for integrating traceability, verifiable reputation, and automated smart contracts into a holistic platform that enhances market efficiency and equity for Sri Lankan farmers?

RQ4: What is the measurable economic impact of such a platform on key performance indicators, such as the farmer's share of the final price and reduction in payment delays, when evaluated in a controlled pilot study?

1.4 Research Problem

The Sri Lankan fruit supply chain is dominated by an inefficient and opaque intermediary system that creates significant value loss, suppresses farmer income, and inflates consumer prices. Smallholder farmers lack the market power and transparent information needed to secure fair prices and are often subject to exploitative practices, including delayed payments and credit-bound relationships. This traditional structure fails to reward quality and perpetuates high post-harvest losses. There is currently no trusted, transparent, and accessible mechanism that enables farmers and buyers to connect and transact directly, based on verifiable performance and quality metrics.

2 OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design, develop, and evaluate a blockchain-powered direct marketplace that leverages a verifiable reputation system to disintermediate the traditional fruit supply chain. This will enhance transparency and ensure fair pricing to improve the livelihoods of Sri Lankan smallholder farmers, while also enabling consumers to access higher-quality, fresher fruits at more affordable prices.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives have been identified:

1. To conduct a detailed analysis of the existing intermediary-led market structure, identifying the specific power dynamics, economic inefficiencies, and trust mechanisms that disadvantage farmers and buyers.
2. To design a secure, private, and scalable system architecture using Hyperledger Fabric, detailing the process for end-to-end traceability and the creation of a verifiable on-chain reputation for farmers using Decentralized Identifiers (DIDs).
3. To develop a functional prototype of the marketplace, including the implementation of an intelligent matching algorithm and smart contracts that automate direct sales agreements, enforce quality-based pricing, and facilitate secure, timely payments between verified participants.
4. To evaluate the feasibility, usability, and potential economic impact of the prototype through a controlled pilot study, measuring key performance indicators (KPIs) such as the change in farmer's share of the final price, reduction in payment delays, and overall user trust in the direct marketplace model.

3 Methodology

3.1 Research Approach and Design

This study will adopt a mixed-methods research design, combining qualitative inquiry with a design science research (DSR) approach for system development and evaluation. This methodology is well-suited for solving real-world problems through the creation and validation of a novel IT artifact.[1] The research will be conducted in five distinct phases, providing a structured path from problem understanding to solution validation.

Phase 1 & 2: Problem Identification and Objective Definition

This foundational phase is dedicated to deeply understanding the problem domain and defining the objectives for the solution. This will be achieved by gathering detailed requirements from key stakeholders.

- **Data Collection:** The primary data collection method will be semi-structured interviews. Separate interview protocols will be designed for three key stakeholder groups to ensure a holistic understanding of the ecosystem:
    - **Smallholder Fruit Farmers:** To understand their current financing methods, relationships with intermediaries, pricing mechanisms, payment cycles, challenges in accessing capital, and willingness to adopt new technologies. (See Appendix A for sample questionnaire).
    - **Buyers (Exporters, Processors, Supermarket Procurement Managers):** To identify their procurement challenges, quality requirements, interest in direct sourcing, and the perceived benefits and risks of engaging directly with farmers. (See Appendix B for sample questionnaire).
    - **Intermediaries (Collectors, Traders at DECs):** To understand their business model, costs, perceived value, and operational challenges within the current supply chain.

- **Data Analysis:** The interview transcripts will be analyzed using thematic analysis to identify recurring themes, pain points, and critical success factors for the proposed system. The findings from this phase will directly inform the system's functional and non-functional requirements, ensuring the design addresses the real-world needs of its intended users.

3.2 Phase 2: System Design and Architecture

This phase follows the design science research paradigm to build the technological artifact. It involves defining the complete technical blueprint for the platform.

3.2.1 Technology Stack and Justification

The technology stack is chosen to ensure security, privacy, scalability, and ease of development for an enterprise-grade supply chain platform.

Table 4: Technology Stack Justification

Component	Technology	Justification
Blockchain Platform	Hyperledger Fabric	Enterprise-Grade & Permissioned: Unlike public chains (Solana, Polygon), Fabric is a permissioned network, meaning only verified participants can join. This is essential for a B2B context to ensure accountability and data privacy. Confidentiality: Fabric's "channel" and "private data collection" features allow specific parties to transact confidentially, protecting sensitive business information like pricing from competitors on the same network.
Smart Contracts	Go (Golang)	Performance & Security: Go is a statically typed, compiled language known for its high performance and concurrency, making it ideal for writing secure and efficient smart contracts (chaincode) that handle complex business logic.
Backend API	Node.js with Express.js	Mature SDK & Asynchronous I/O: The official Hyperledger Fabric SDK for Node.js is robust and well-supported. Node.js's non-blocking, event-driven architecture is highly efficient for handling API requests between the user applications and the blockchain network.

Frontend Application	React Native (Mobile) & React.js (Web)	Cross-Platform & User-Centric: React Native allows for the development of a single mobile app for both Android and iOS, which is crucial for reaching a wide farmer base in Sri Lanka. React.js is a powerful standard for building the responsive web portals needed by buyers and administrators. This approach prioritizes a simple, accessible user experience to address potential digital literacy gaps.
State Database	CouchDB	Rich Query Capabilities: When used as the state database for Fabric peers, CouchDB stores data in JSON format. This enables complex, real-time queries on the current state of the ledger (e.g., finding all farmers with a certain reputation score), which is not possible with the default LevelDB.
Off-Chain Storage	IPFS (InterPlanetary File System)	Efficiency & Verifiability: Storing large files (e.g., quality certificates, images) directly on the blockchain is inefficient. IPFS provides decentralized storage. We will store the file on IPFS and record only its immutable hash on the blockchain, ensuring the data is tamper-evident without bloating the ledger.

3.2.2 System Architecture Design

The system will be designed with a multi-layered architecture, including a presentation layer (frontend apps), a business logic layer (backend API and smart contracts), and a data persistence layer (blockchain ledger and off-chain storage), to ensure modularity and scalability.

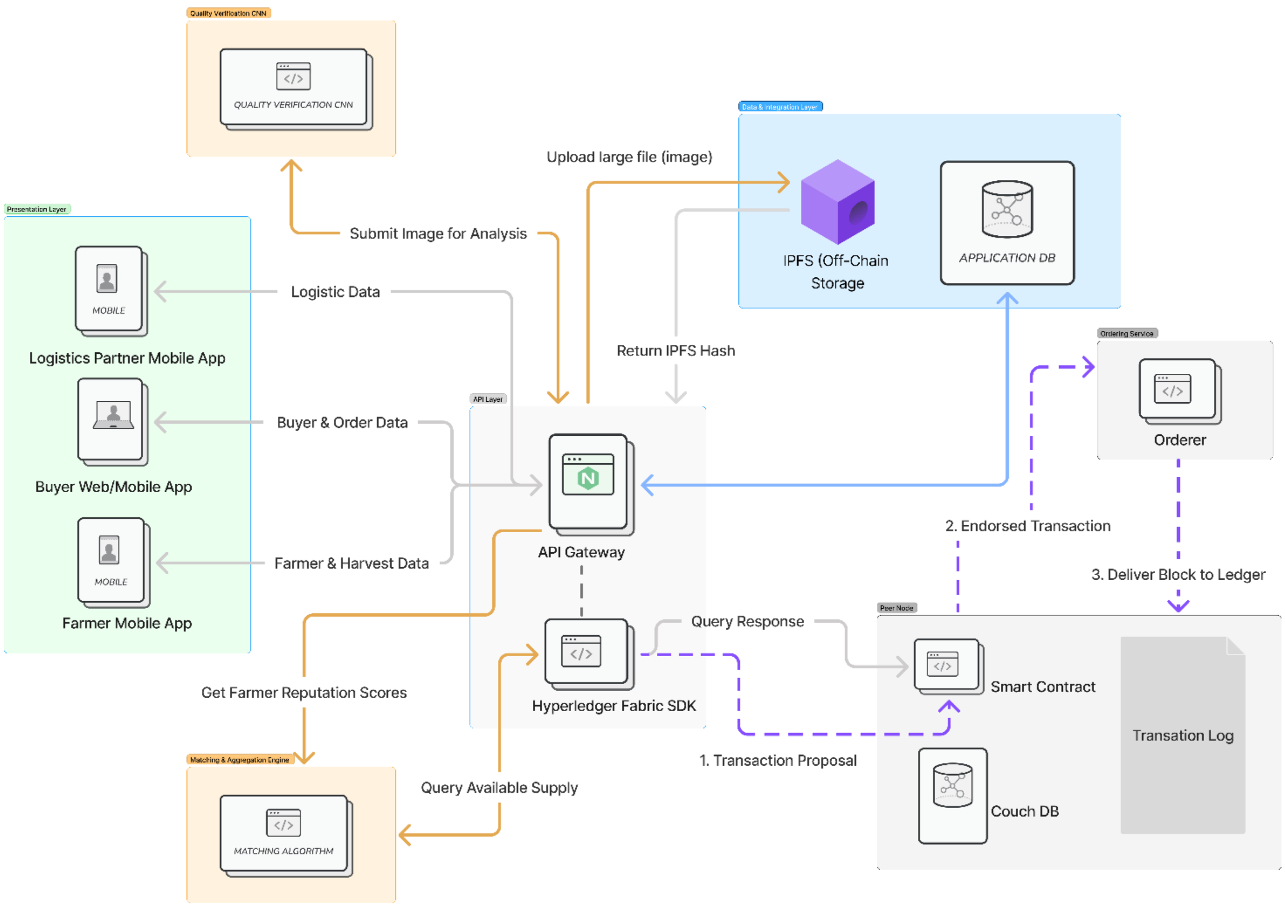


Figure 2: System Architecture Diagram

3.2.3 Intelligent Matching and Aggregation Algorithm

The core of the marketplace is an Automated Order Fulfillment and Aggregation Engine. This engine moves beyond simple listings to solve the complex problem of sourcing a specific quantity of produce from multiple, fragmented smallholder farmers. It functions as a multi-objective optimization system designed to find the optimal combination of farmers to fulfill a buyer's order efficiently and reliably.

Algorithmic Steps:

- **Step 1: Data Ingestion:** The algorithm is triggered by a buyer's purchase order, which specifies *productID, quantityNeeded, qualityGrade, deliveryDate*, and *deliveryLocation*. The system simultaneously queries the blockchain for all farmers who have registered an available supply matching these primary criteria. Each farmer's record includes their *farmerID, availableQuantity, askingPricePerKg, farmLocation*, and their current on-chain *reputationScore*.
- **Step 2: Initial Filtering:** A pool of eligible farmers is created by applying hard constraints from the buyer's order. This initial filtering is a simple database query on the blockchain's world state (CouchDB).
- **Step 3: Multi-Factor Scoring and Ranking:** This is the central logic of the engine. For each farmer in the eligible pool, a Match Score is calculated to determine their priority in the fulfillment process. The score is a weighted sum of normalized factors:

$$Match\_Score=(wrep \times Reputation\_Score)+(wprice \times Price\_Score)+(wloc \times Location\_Score)$$

  - **Weights (w):** These are configurable parameters (e.g., wrep=0.5,wprice=0.3,wloc=0.2) that allow the system to prioritize different factors. For high-value export orders, the reputation weight (wrep) might be increased.

  - **Reputation Score:** This is the farmer's on-chain reputation score (normalized 0-1), derived from their history of successful deliveries, consistent quality grades, and positive buyer feedback. This is the most critical factor for ensuring trust.

  - **Price Score:** This score is normalized to reward lower prices. A simple formula is:

$$Price\_Score=1-(farmer\_price/buyer\_max\_price)$$

- **Location Score:** This score promotes logistical efficiency. It is calculated based on the farmer's geographical proximity to other high-scoring farmers in the eligible pool. A clustering algorithm (e.g., K-Means) can be run on the farm locations, and farmers in denser clusters receive a higher score.

After calculating the *Match_Score* for every eligible farmer, the list is sorted in descending order.

- **Step 4: The Aggregation Loop:** The system iterates through the sorted list to build the fulfillment package using a greedy algorithm approach

- **Step 5: Proposal and Execution:** The resulting fulfillment_group is presented to the buyer as a complete fulfillment plan. Upon confirmation, the system deploys a single Master Smart Sales Agreement to the blockchain. This master contract manages the overall order and creates individual, linked sub-contracts for each farmer in the group, handling payment escrow and automated settlement for each one upon successful delivery verification.

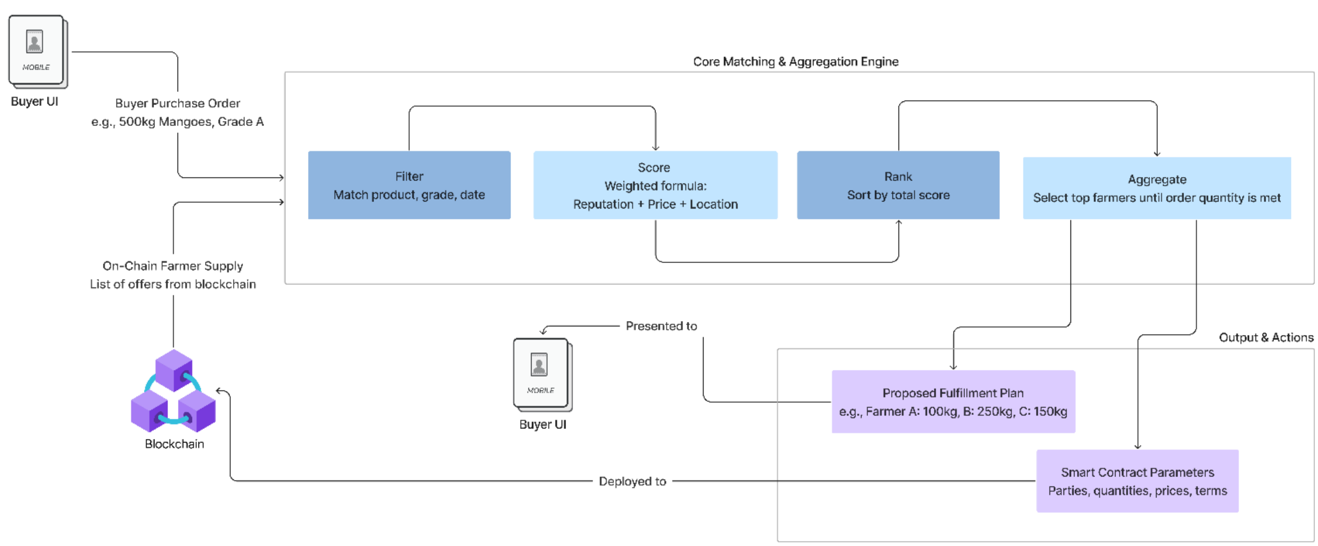


Figure 3: Matching Algorithm Architecture

3.2.4 Project Timeline and Gantt Chart

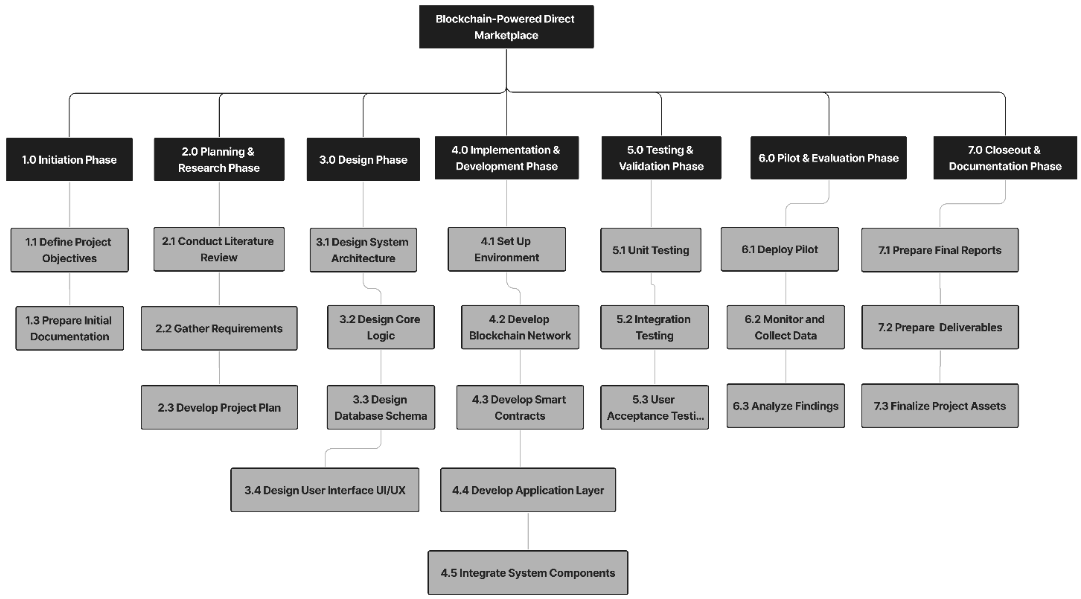


Figure 4: Work Breakdown Chart

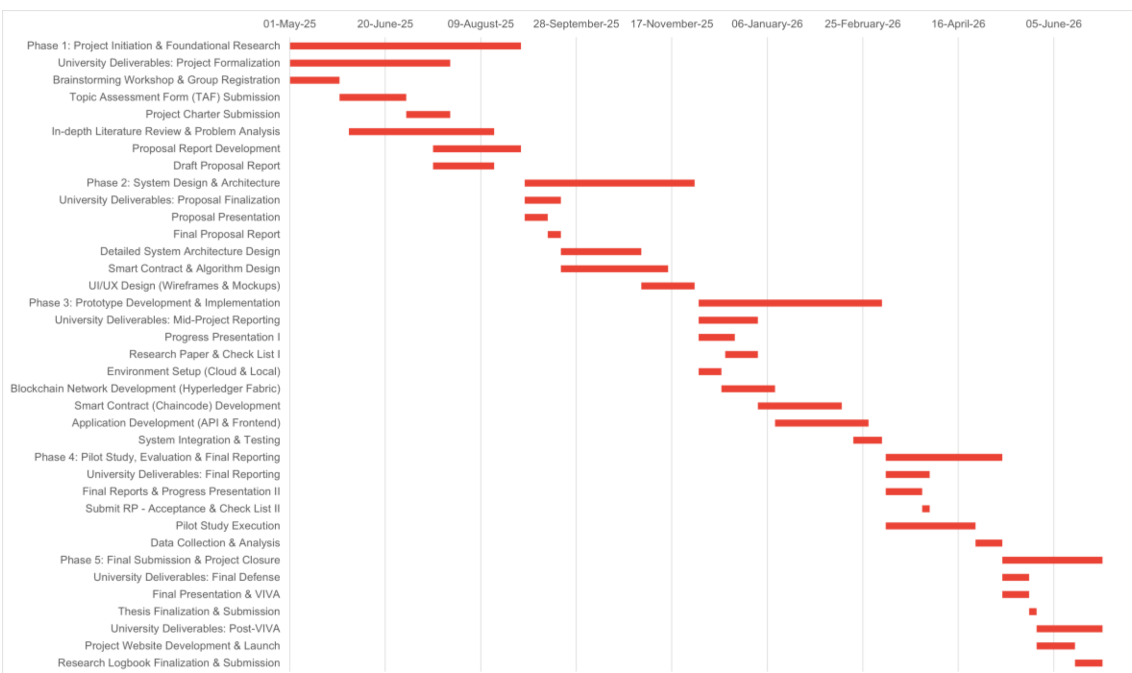


Figure 5: Gantt Chart for Project Timeline (Jan 2025 – May 2026)

4 DESCRIPTION OF PERSONAL AND FACILITIES

4.1 Personal

This research is the individual component of a larger group project and will be conducted by a final-year undergraduate student from the Department of Information Technology at the Sri Lanka Institute of Information Technology. As the lead for this component, I possess a specialization in blockchain architecture and smart contract development, with proficiency in Go, JavaScript, and the Hyperledger Fabric framework. I will be responsible for the overall system design, technical implementation, and evaluation of the blockchain-powered marketplace.

This individual research will be carried out under the direct supervision of ****, who has extensive expertise in distributed systems and their application in enterprise environments. Regular consultations and guidance from my supervisor will ensure the academic rigor and technical soundness of the project. I will also collaborate closely with my peers who are handling the other three research components (AI Demand Forecasting, Spoilage-Reduction Logistics, and Quality Assurance) to ensure seamless integration of our respective modules into the final, cohesive system.

4.2 Facilities

The research will be conducted utilizing the state-of-the-art computing laboratories and resources provided by the Sri Lanka Institute of Information Technology. This includes access to high-performance workstations for system development and simulation, university-provided cloud computing resources for deploying the blockchain network, and subscriptions to academic databases (e.g., IEEE Xplore, ACM Digital Library) for the literature review. The university's network and facilities will provide the necessary infrastructure to support all phases of the research, from development to testing and final analysis.

5 Project Requirements

This section outlines the functional and non-functional requirements for the blockchain-powered marketplace component. It also details the personal requirements needed for the research and the expected test cases that will be used to validate the system's performance and correctness.

5.1 Functional Requirements

- **Secure User Onboarding and Identity Management** - The system must provide a secure registration process for all participants (farmers, buyers, etc.). Upon verification by a trusted administrator, the system must generate a unique, non-custodial **Decentralized Identifier (DID)** for each user on the blockchain, serving as their sovereign digital identity.

- **On-Chain Reputation Management** - The system must automatically calculate and update a user's reputation score after each completed transaction. This score, based on verifiable credentials such as on-time delivery and quality grades, must be immutably recorded on the blockchain and linked to the user's DID.

- **Farmer Supply Listing** - The farmer-facing application must allow users to list their predicted future harvest, including details such as fruit type, expected quantity, quality grade, and available dates. Farmers must have the option to list their supply individually or pool it as part of a registered digital cooperative.

- **Buyer Order Placement and Matching** - The buyer-facing portal must allow users to create detailed purchase orders. The system's **Intelligent Matching & Aggregation Engine** must be able to process these orders, filtering and ranking farmers based on reputation, price, and location, and aggregating supply from multiple farmers to fulfill a single large order.

- **Smart Contract Automation** - For every matched trade, the system must automatically deploy a **Smart Sales Agreement** to the blockchain. This contract must be able to securely hold the buyer's payment in **escrow** and automatically release the funds to the respective farmers upon verifiable confirmation of successful delivery.

- **Transaction Transparency and Traceability** - All relevant parties to a transaction must be able to view its current status (e.g., "Awaiting Delivery," "Payment in Escrow," "Completed") by querying the blockchain ledger. This ensures a shared, single source of truth.

- **Admin Dashboard and Governance** - A secure web-based dashboard must be available for administrators. This dashboard must provide functionalities for verifying and managing network participants, monitoring the health of the blockchain network, and resolving disputes.

5.2 Non-Functional Requirements

- **Security**
  Since the system handles financial transactions and sensitive identity data, security is the highest priority. The architecture must implement end-to-end encryption, secure private key management for users, and role-based access control. Hyperledger Fabric's channel architecture will be used to ensure the privacy of confidential transaction data.

- **Scalability**
  The platform must be designed to scale efficiently as the number of farmers, buyers, and transactions grows. The Hyperledger Fabric network will be architected in a modular way, allowing for the addition of new peer nodes and organizations without disrupting the existing network.

- **Usability**
  The mobile application for farmers must be extremely simple, intuitive, and designed for users with varying levels of digital literacy. The interface should provide clear visual cues and require minimal text input. The buyer and admin portals must be powerful yet user-friendly, with clear dashboards and reporting features.

- **Reliability & Availability**
  The blockchain network must be designed for high availability (e.g., 99.9% uptime) to be a trustworthy platform for commercial trade. This will be achieved through redundant peer nodes and a robust cloud hosting infrastructure.

- **Performance**
  Transaction latency must be low to ensure good user experience. Read queries (e.g., fetching a reputation score) should be completed in under 2 seconds, and write transactions (e.g., confirming a delivery) should be committed to the ledger within 5-10 seconds.

- **Immutability**
  A core requirement of the blockchain layer. Once a transaction (e.g., a completed sale, an update to a reputation score) is written to the ledger, it must be cryptographically secured and impossible to alter or delete, ensuring a permanent and trustworthy audit trail.

- **Transparency**
  While ensuring the confidentiality of specific trade details, the system must provide transparency into the processes that govern it. For example, the logic of the reputation scoring algorithm and the conditions for smart contract settlement will be open and auditable by authorized participants**.**

5.3 Expected Test Cases

Table 5:Test case 1 - User Onboarding & DID Creation

Objective	To verify that a new, verified user is correctly onboarded and a unique Decentralized Identifier (DID) is created for them on the blockchain.
Test Case	Create a new farmer account and have it approved by an admin. Check the blockchain ledger to confirm that a new DID has been generated and is linked to the farmer's profile.
Expected Outcome	The system should successfully create a unique DID for the user, and this identity should be queryable on the ledger.

Table 6: Test case 2 - Smart Contract Execution

Objective	To validate the full lifecycle of the Smart Sales Agreement, including payment escrow and automated settlement.
Test Case	Initiate a trade between a buyer and a farmer. Verify that the buyer's funds are correctly transferred to and held by the smart contract's escrow. Simulate a successful delivery confirmation and verify that the funds are automatically and correctly released to the farmer's wallet.
Expected Outcome	The smart contract should manage the escrow and settlement process automatically and accurately without any manual intervention.

Table 7: Test case 3 - Reputation Score Update

Objective	To ensure that a completed transaction correctly triggers an update to the farmer's on-chain reputation score.
Test Case	Complete a transaction for a farmer with an initial reputation score of 80. After successful completion, query the reputation management smart contract.
Expected Outcome	The farmer's reputation score should increase based on the predefined scoring algorithm, and a new Verifiable Credential should be linked to their DID.

Table 8: Test case 4 - Matching & Aggregation Algorithm

Objective	To test the accuracy and efficiency of the Intelligent Matching & Aggregation Algorithm.
Test Case	Create a large purchase order that requires supply from multiple farmers. Provide a pool of farmers with varying reputation scores, prices, and locations.
Expected Outcome	The system should correctly propose a fulfillment plan that prioritizes high-reputation farmers and aggregates the required quantity efficiently.

Table 9: Test case 5 - Data Privacy on Channels

Objective	To verify that confidential transaction data is kept private between the transacting parties.
Test Case	Use a load testing tool to simulate hundreds of concurrent users performing actions such as placing orders, confirming deliveries, and querying the ledger.
Expected Outcome	The system should remain stable, with transaction latency and query response times staying within the predefined performance targets.

6 Commercialization

6.1 Vision and Market Opportunity

The vision for this specific component is to become the definitive digital trust layer for Sri Lanka's fruit supply chain. The immediate market opportunity is to solve the most fundamental problem plaguing the sector: a lack of trust and transparency between farmers and buyers.

The blockchain marketplace is the essential first step, directly tackling the core issues of unfair pricing, payment delays, and information asymmetry that lead to significant value loss [26]. By establishing this trusted transaction platform, we create the necessary foundation upon which other value-added services can be integrated later. The global validation for such platforms is strong, with significant venture capital flowing into agri-food tech solutions that improve supply chain integrity [6].

6.2 Target Market

The platform will utilize a B2B model with a multi-sided market strategy:

- **Primary Customers (Buyers):** Exporters, processors, and supermarket chains are the primary paying customers. They gain the most from the platform's core offerings: access to a pool of verified farmers, reduced counterparty risk, and verifiable traceability data for their products.[12][27]
- **Platform Users (Farmers):** Smallholder farmers are the primary beneficiaries and the key to building a robust supply network. To ensure widespread adoption, access for farmers will be free, addressing a common barrier to entry for new technologies in rural areas.[28]
- **Ecosystem Partners:** The marketplace is designed to be a hub. Financial institutions, for instance, can directly leverage the on-chain reputation data for risk assessment in loan applications, making the marketplace their primary source for verified performance history.

6.3 Business Model and Revenue Streams

The revenue model is designed to be self-sustaining by capturing a fraction of the value it creates through secure and efficient transactions.

1. **Transaction Commission Fee:** The main revenue stream will be a commission fee charged to the buyer for each completed transaction facilitated by the marketplace's smart contracts. This directly monetizes the platform's primary function: enabling trusted trade.
2. **Premium Subscription Tiers (SaaS):** For high-volume buyers, a subscription model will offer premium features derived directly from the marketplace's data and functionality. A similar model has been successfully used by international platforms like Australia's AgriDigital [16]. Features include:
   - **Advanced Data Access:** API access for buyers to integrate marketplace data into their own procurement software.

- **Enhanced Reputation Analytics:** Detailed dashboards on farmer performance metrics drawn from their immutable on-chain history.
- **Priority Matching:** Priority placement within the platform's intelligent matching and aggregation algorithm.

3. **Data Monetization:** Anonymized and aggregated data on transaction volumes and pricing, all verified by the blockchain ledger, will be sold as high-integrity market intelligence reports to government bodies [10] and international agencies [8].

6.4 Go-to-Market Strategy

The rollout will focus on establishing the marketplace as the central, indispensable platform for trade.

- **Phase 1:** Establish the Trust Engine (Year 1): The pilot will focus exclusively on validating the core marketplace functions: farmer onboarding, creation of DIDs, direct matching, and successful execution of smart contract-based sales. This "crossing the chasm" strategy focuses on proving clear ROI in a niche market before scaling [30].
- **Phase 2:** Scale the Network (Year 2-3): The focus will be on aggressively onboarding farmers and buyers. We will partner with Farmer Producer Organizations and use agricultural extension networks to drive adoption of the core platform [28].
- **Phase 3:** Open for Integration (Year 4 onwards): Once the marketplace is established, we will open APIs for other services to integrate. This positions the marketplace as the core platform of a wider agri-tech ecosystem.

6.5 Competitive Advantage

The competitive advantage of the blockchain marketplace component is uniquely defensible:

- **The Verifiable Trust Network:** This is the core moat. The on-chain reputation system using DIDs and VCs is the platform's unique selling proposition. It directly addresses the market failures and conflicts caused by information asymmetry [9].
- **Data Integrity:** Unlike traditional platforms, the data on pricing and transactions on our marketplace is transparent and tamper-evident by design. This integrity is a powerful asset for all stakeholders.

- **Intelligent Aggregation:** The marketplace's built-in matching and aggregation algorithm is a key feature that solves the real-world problem of sourcing large orders from many small farmers.
- **Alignment with National Policy:** The platform directly supports Sri Lanka's Agriculture Sector Modernization Project, which aims to improve market linkages and enhance export competitiveness [5][11].

7 Budget and Budget Justification

This is a projected budget for the development and pilot phase of this individual research component

Table 10: Budget Table

Item	Estimated Cost (LKR)
Cloud Services	25000.00
Software & Tools	5000.00
Internet cost	5000.00
Field Research & Miscellaneous	15000.00
Total Estimated Cost	50000.00

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9 Appendices

9.1 Appendix A: Sample Farmer Interview Questionnaire

Section 1: Demographic and Farm Information

1. Name of Farmer:
2. Location (District, Village):
3. Age:
4. Total farm size (in acres):
5. Primary fruit crops cultivated:
6. How many years have you been farming?

Section 2: Current Marketing and Sales Practices

7. To whom do you primarily sell your harvest? (e.g., Local collector, Trader at Economic Center, Supermarket, Exporter, Direct to consumer)
8. How is the price for your produce determined? Who sets the price?
9. How and when do you receive payment for your sales? Are there often delays?
10. What are your biggest challenges when selling your produce? (e.g., low prices, transport, finding buyers)

Section 3: Relationship with Intermediaries

11. Do you have a regular collector or trader you sell to?
12. Does this person provide you with any other services, such as loans, seeds, or fertilizer?
13. Do you feel you get a fair price for your produce? Why or why not?
14. Have you ever tried to sell your produce directly to a larger market or buyer? What was your experience?

Section 4: Technology Adoption and Perceptions

15. Do you own a smartphone? Do you use it for farming-related activities?
16. How comfortable are you with using mobile applications?
17. If a system existed that allowed you to see prices from many buyers and sell directly to them, would you be interested?
18. What would be your main concerns about using such a system? (e.g., trust, complexity, transport)
19. What features would be most important to you in a digital platform for selling your produce?

9.2 Appendix B: Sample Buyer Interview Questionnaire

Section 1: Company and Procurement Information

1. Name of Company & Your Role:
2. Type of Business (e.g., Exporter, Processor, Supermarket Chain):
3. What are the primary fruits you procure from Sri Lanka?
4. What are your annual volume requirements for these fruits?

Section 2: Current Sourcing and Supplier Management

5. How do you currently source your produce? (e.g., Direct from large farms, through contract suppliers, from Economic Centers, from collectors)
6. What are your biggest challenges in procuring produce of the required quality and quantity?
7. How do you ensure the quality and safety of the produce you buy? What is your process for rejecting a shipment?
8. What are the main risks you face in your current supply chain? (e.g., price volatility, supply inconsistency, quality issues, lack of traceability)

Section 3: Perceptions of Direct Sourcing and Technology

9. Have you considered or attempted direct sourcing from smallholder farmers? What were the barriers or outcomes?
10. What would be the main benefits for your business if you could source directly from a large, verified pool of high-quality farmers?
11. How important is traceability ("farm-to-fork") for your business and your end-customers?
12. Would a system that provides a verifiable, unchangeable record of a farmer's past performance (e.g., quality grades, on-time deliveries) be valuable to you?
13. What features would be essential for you in a digital platform for direct procurement? (e.g., automated order matching, quality verification, secure payments, logistics coordination)